

Ever since I started watching the NBA, I always wondered why teams shoot so many three pointers when it is the most difficult shot to make. On top of that, it makes the game less attractive to watch in my opinion.

When I started looking into the topic, I found that this is a common observation. A lot of fans noticed a clear tendency towards the 3-pointer starting from the 2015-16 season, when the Golden State Warriors changed the NBA forever with their 3-point shooting. With the "Splash Brothers" — Steph Curry and Klay Thompson — the Warriors shook the whole league with their offensive strategy and finished with a historic 73-9 record.

This made me question whether the 3-pointer is truly a game decider. My first instinct was yes, because we can see how the whole league has adapted : teams now build their roster around 3-point shooting, and a player who cannot shoot from range is considered incomplete.

But is it really that simple ?

Does shooting more 3-pointers lead to more wins, or is it about shooting them well ?

Data :

- Source : NBA official stats via the `nba_api` Python library
- 30 NBA teams × 9 seasons (2015-16 to 2023-24)
- 270 observations in total
- Regular season only

Results :

	Correlation with wins (%)
3-Point Attempts	0.178
3-Point Shooting %	0.575

→ Shooting efficiency from 3 is 3 times more correlated with wins than the volume of attempts. It is not about how many you shoot — it is about how many you make.

Conclusion :

The data confirms that 3-point shooting efficiency is a strong predictor of winning in the NBA. But the story does not stop there.

Looking at the evolution over 9 seasons, we can see that teams are attempting more and more 3-pointers every year, while the league average shooting percentage has remained relatively stable. This means that the players entering the league today are simply better 3-point shooters than the previous generation — shooting from range has become a fundamental skill, just like dribbling or passing.

This leads to an interesting strategic conclusion : since the league as a whole is getting better at shooting the 3, the optimal strategy is no longer just to shoot it well — it is to shoot it well **and** at a high volume. A team that combines both a high 3-point attempt rate and a high shooting percentage has a significant competitive advantage over teams that rely on a more traditional style of play.

In short, the Warriors did not just win championships — they changed the way the game is played forever. The 3-pointer is no longer a bonus option. It is the foundation of modern NBA basketball.

How to run the program :

1 - Install the dependencies :

```
pip install pandas matplotlib scipy numpy nba_api
```

2 - Run the code